第32套.

1. (1). 设长.h.

$h^2+(2\pi r)^2=L^2$

$V_0=h\cdot\pi r^2$.

$=\pi r^2\sqrt{L^2-4\pi^2 r^2}$.

$\left(\frac{a^2+b^2+c^2}{3}\right)^{\frac{1}{2}}\geqslant\frac{a+b+c}{3}\geqslant(a\cdot b\cdot c)^{\frac{1}{3}}$

$=\frac{1}{2\pi}\sqrt{2}\pi\cdot r\sqrt{2}\pi r\cdot\sqrt{L^2-4\pi^2r^2}$.

当$\sqrt{2}\pi r=\sqrt{L^2-4\pi^2r^2}$时.有最值.

$V_m=\frac{\sqrt{3}}{18\pi}L^3$

$\Rightarrow 2\pi^2r^2=L^2-4\pi^2r^2$.

$\Rightarrow \theta=\arccos\frac{\sqrt{6}}{3}$,

(2).设长为h.截面周长.c.

$$\begin{cases}V_0=sh\\c^2+h^2=L^2\\s\leqslant\frac{c^2}{4\pi}\end{cases}$$
.(当且仅当圆时有等号).

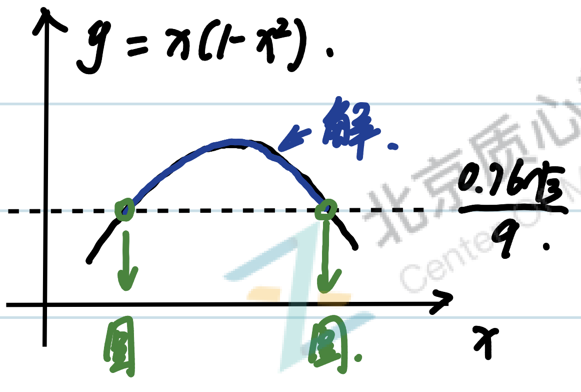
$$\Rightarrow\begin{cases}L^2-h^2\geqslant4\pi s.\\V_0=s\cdot h.\end{cases}$$

$\Rightarrow h(L^2-h^2)\geqslant4\pi\cdot0.38\frac{\sqrt{3}}{18\pi}L^3$.

令$x=\frac{h}{L}$. 有

$\Rightarrow x(1-x^2)\geqslant\frac{0.76}{9}\sqrt{3}$

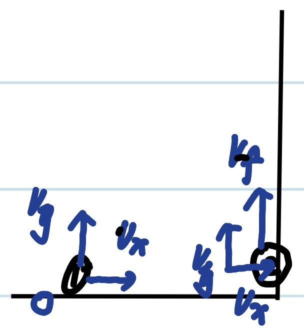
当且仅当圆有等号.



由.计算器得.$x_1=0.15$

$x_2=0.92$

2.



①

$I_x=mv_x$. $I_y=mv_y$. $I_x^2+I_y^2\leqslant I_0^2$.

$\Rightarrow(mv_x)^2+(mv_y)^2\leqslant mv_0^2$.

$aI_y\geqslant I_x\Rightarrow v_y\geqslant\frac{1}{a}v_x$.

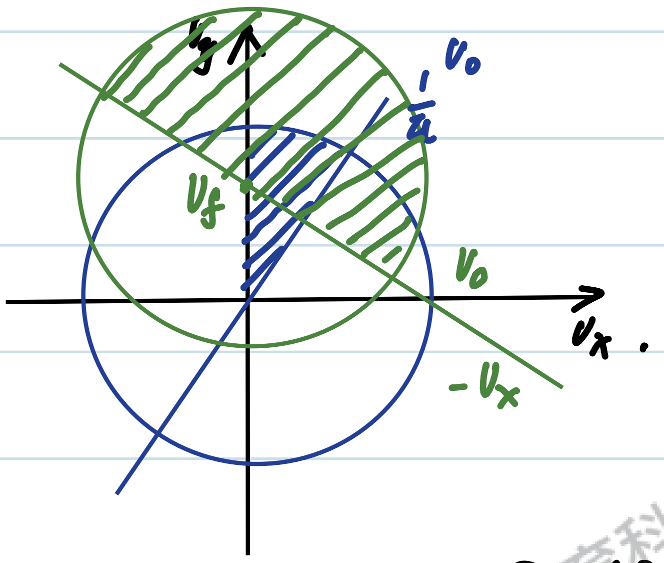
②. $I_x=mv_x$. $I_y=m(v_f-v_y)$. $I_x^2+I_y^2\leqslant I_0^2$

$\Rightarrow v_x^2+(v_y-v_f)^2\leqslant v_0^2$.

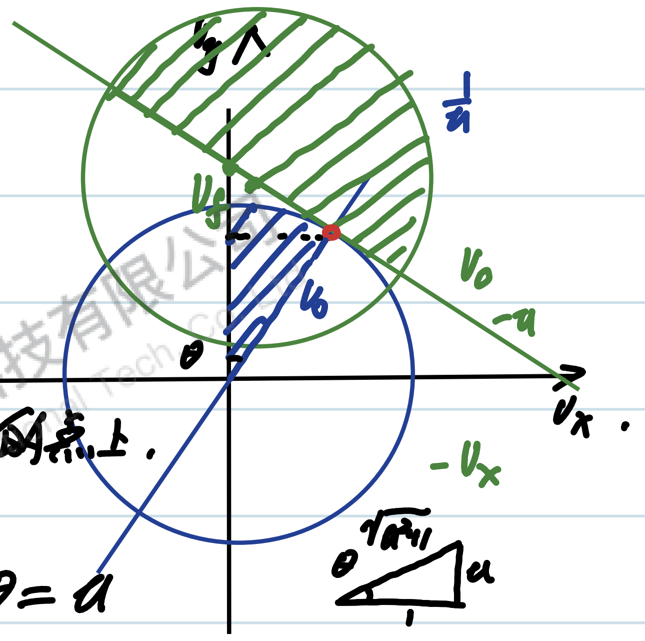
$I_y\leqslant aI_x\Rightarrow$. $v_f-v_y\leqslant av_x$.

满足①.②的 v_f 的最值.

物理考虑临界.相圆.(线性规划)



重叠部分.是解.



极值是刚好相切.最边缘的点上.

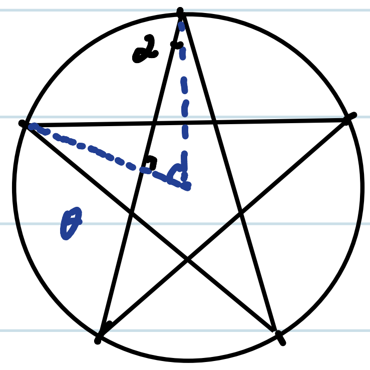
$\Rightarrow$. $v_f\cos\theta=v_0$. $\tan\theta=a$

$\Rightarrow v_f=\sqrt{a^2+1}\,v_0$.

$2gy=v_f^2$.

$h=l+y=l+\frac{v_f^2}{2g}=l+\frac{(a^2+1)v_0^2}{2g}$.

(3).



$\theta=\frac{360^\circ}{5}=72^\circ$. $\alpha=18^\circ$.

设纸面内速度.v_{xy}.

垂直纸面速度.v_z.

$$\begin{cases} I_N = 2mv_{xy}\cdot\cos18^\circ \\ I_f = 2mv_z. \end{cases}$$

$$\begin{cases} uI_N \geq I_f. \\ I_N^2+I_f^2 \leq I_0^2. \end{cases}$$

$$\Rightarrow. \begin{cases} u\cdot v_{xy}\cos18^\circ \geq v_z. ① \\ v_{xy}^2\cos^2 18^\circ + v_z^2 \leq \left(\frac{v_0}{2}\right)^2 ② . \end{cases}$$

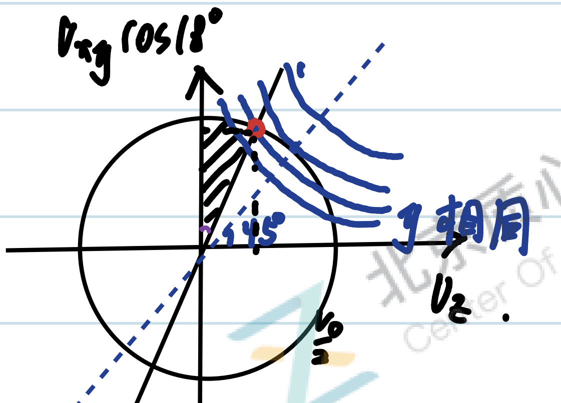
室中飞行.

z: $gt = 2v_z$

xy: $v_{xy}\cdot t = 2r\cdot\cos18^\circ$.

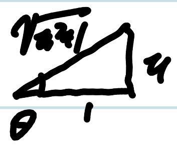
$$\Rightarrow. \ r = \frac{(v_{xy}\cdot\cos18^\circ)\cdot v_z}{g\cos^2 18^\circ}.$$

r 最大值. 求 $y = (v_{xy}\cos18^\circ)\cdot v_z$ 的最值.


$v_{xy}\cos18^\circ$
45°
g 相同
v_z.
$\frac{v_0}{2}$

取边缘点时. 有 y_{max}.

$\tan\theta = a$.


$\sqrt{a^2+1}$
a
θ
1

$$v_z = \frac{v_0}{2}\sin\theta = \frac{v_0}{2}\frac{a}{\sqrt{1+a^2}}$$

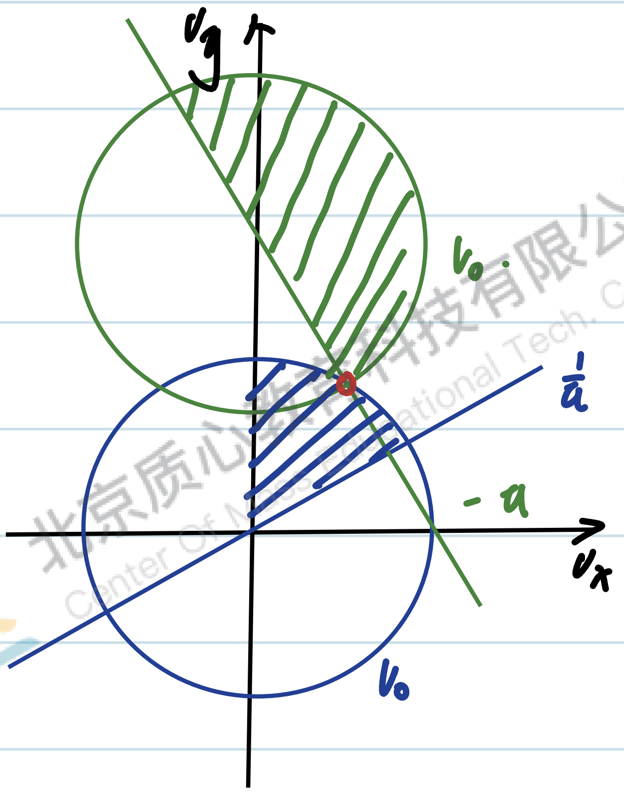
$$v_{xy}\cdot\cos18^\circ = \frac{v_0}{2}\cos\theta = \frac{v_0}{2}\cdot\frac{1}{\sqrt{1+a^2}}.$$

$$r = \left(\frac{v_0}{2}\right)^2\frac{u}{1+a^2}\cdot\frac{1}{\cos^2 18^\circ}\frac{1}{g}$$

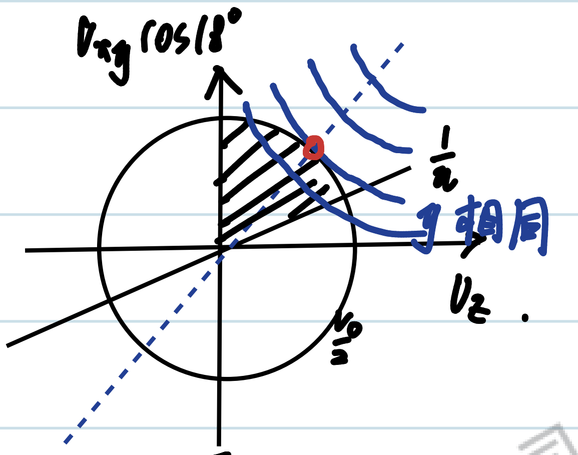
以上. $u<1$ 的结果.

若 $a>1$. 则:

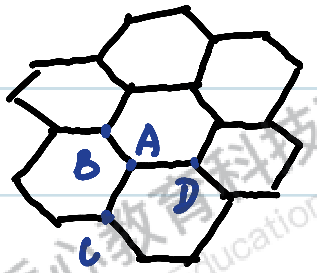
(1).


v_y
v_0
$\frac{1}{a}$
$-a$
v_x.
v_0

(2).


$v_{xy}\cos18^\circ$
$\frac{1}{a}$
g 相同
v_z.
$\frac{v_0}{2}$

3



(1) ① B入 I. ∞出 I.

$B\to A$. $\frac{1}{3}I$.

$A\to C$. $\frac{1}{6}I$.

② ∞入 I. C. 出 I.

$B\to A$. $\frac{1}{6}I$

$A\to C$. $\frac{1}{3}I$.

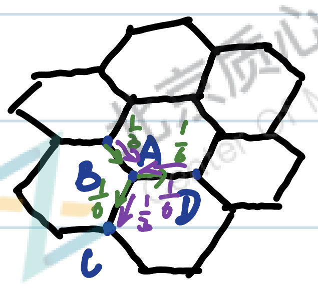
①+②. B入 I. C 出入.

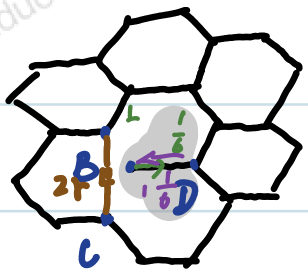
$B\to A$. $\frac{1}{2}I$

$A\to C$ $\frac{1}{2}I$.

$$U_{BC} = \frac{1}{2}I\cdot r + \frac{1}{2}Ir.$$

$$R_{BC} = \frac{U_{BC}}{I} = r.$$

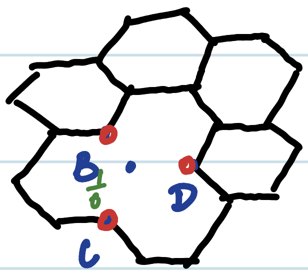

B
A
D
C
$\frac{1}{3}$
$\frac{1}{6}$

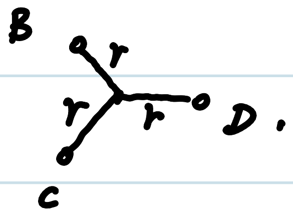

B
D
C
2r
$\frac{1}{3}$

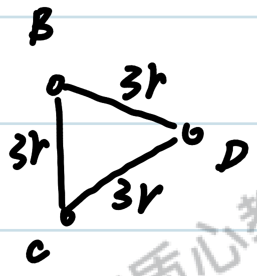
$R_{BC,2} \parallel 2r = R_{BC,1} = r$.

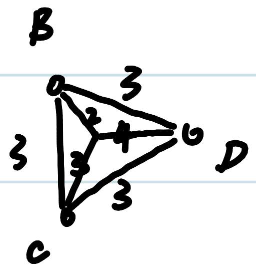
$\Rightarrow R_{BC,2} = 2r$.

(3).



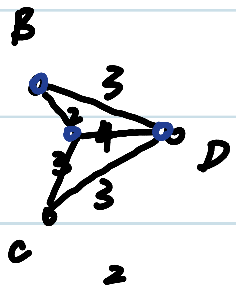




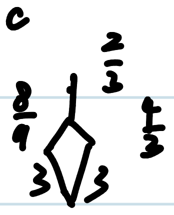


R_{AB}:

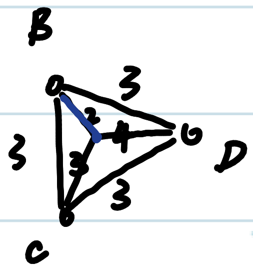
$3 \parallel$



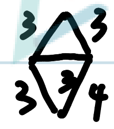
$3 \parallel$


$\frac{2}{3}$
$\frac{8}{9}$
$\frac{4}{3}$
3
3

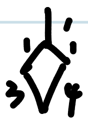
$R = \frac{67}{47} r.$



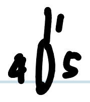
$R_{AB} = 2 \parallel$



$= 2 \parallel$



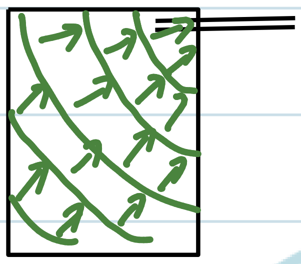
$= 2 \parallel$



$= 2 \parallel \frac{29}{9}$

$= \frac{58}{47} r.$

4.



小孔·缓慢.多孔壁

① 左右有压强差.时也准静态.

②. 左右.只有物质交换.没直接热交换.

$\nabla\cdot\vec{E} = 0.$ $\nabla\cdot\vec{j} = 0$

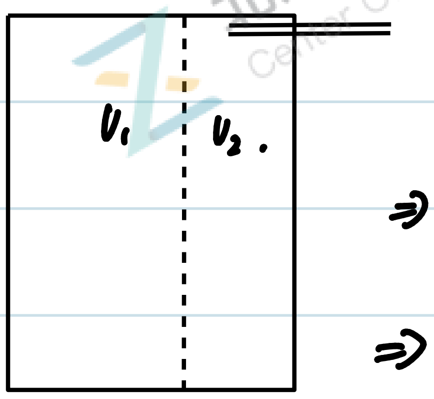
$\nabla^2\varphi = 0.$ $\nabla^2 T = 0.$

唯一性定理.

猜 $T(\vec{r},t) = T(t)$. ⇒ 正确了

⇒ 唯一.

剩下气体绝热



$p_0 V_1^{\gamma} = \frac{p_0}{2} V_0^{\gamma}$

$\Rightarrow \left(\frac{V_1}{V_0}\right) = \left(\frac{1}{2}\right)^{\frac{1}{\gamma}} = 0.61$

$\Rightarrow \nu = \nu\, 0.61.$

(2). $\frac{dV}{dt} = Q = \vec{S}\cdot\vec{v}$. $\frac{dN}{dt} = n\cdot Q = n\vec{S}\cdot\vec{v}$.

$\frac{dN}{dt\cdot dS} = n\cdot\bar{v}_x = \alpha\cdot n\sqrt{\frac{RT}{m}}$

常数.

摩尔数之比. $\frac{d\nu_{N_2}}{d\nu_{O_2}} = \frac{\nu_{N_2}}{\nu_{O_2}}\sqrt{\frac{m_{O_2}}{m_{N_2}}}.$

$\Rightarrow. \ln\frac{\nu'_{N_2}}{\nu_{N_2}} = \sqrt{\frac{8}{7}}\ln\cdot\ln\frac{\nu'_{O_2}}{\nu_{O_2}}.$

压强一半时. $\nu'_{N_2} + \nu'_{O_2} = 0.61\,\nu_0.$

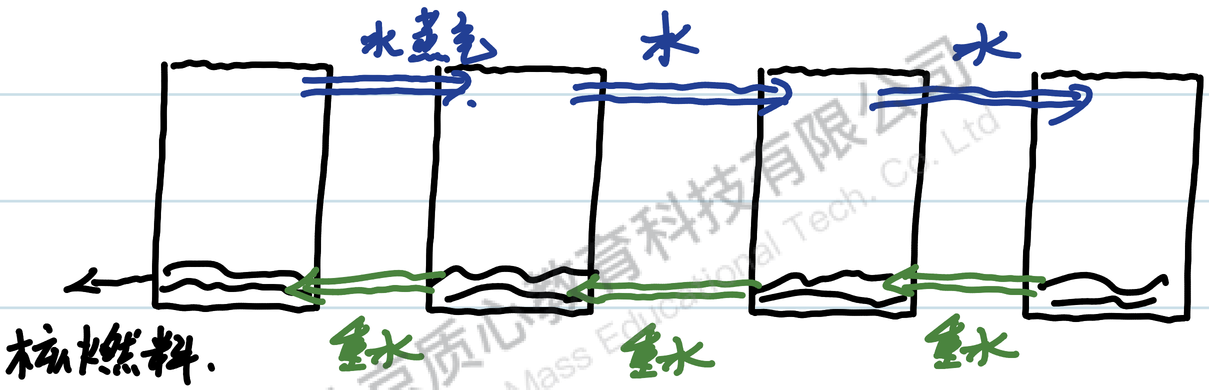
初: $\nu_{O_2} = 0.22\,\nu_0$

$\nu_{N_2} = 0.78\,\nu_0.$

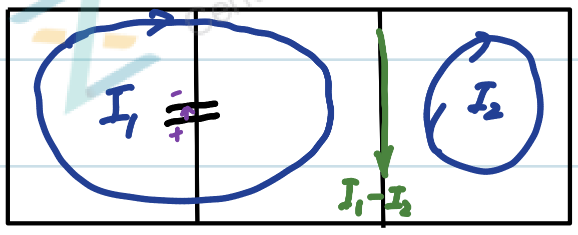
设. $x = \frac{\nu_{N_2}}{\nu_{O_2}}.$

$\Rightarrow \ln\frac{\frac{x}{1+x}0.61}{0.78} = \sqrt{\frac{8}{7}}\ln\frac{\frac{1}{1+x}0.61}{0.22}.$

$\Rightarrow x = 3.43.$


水蒸气
水
水
核燃料.
重水
重水
重水

5.



(1). $\begin{cases} 6I_1 r - I_2 r = 2ka^2 . \\ 4I_2 r - I_1 r = ka^2 . \end{cases}$

$\Rightarrow \begin{cases} I_1 = \frac{9}{23}\frac{ka^2}{r} \\ I_2 = \frac{8}{23}\frac{ka^2}{r} . \end{cases}$

(2). $\frac{Q}{C} + 3I_1 r = \frac{ka^2}{r}$.

$\Rightarrow Q = -\frac{4}{23}\frac{ka^2}{r}$.

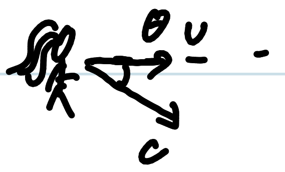
(3). $P = \int \vec{E}_{\text{涡}} \cdot \vec{j} \cdot dV$.

$= 2ka^2 \cdot I_1 + ka^2 \cdot I_2$.

$= \frac{26}{23}\frac{k^2a^4}{r}$

$W = P \cdot t = \frac{26}{23}\frac{k^2a^4}{r} \cdot t$.

6. (1).

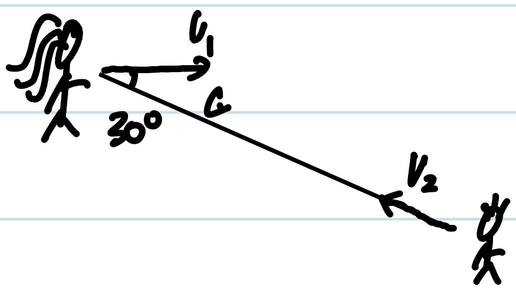


$v_x = \frac{v + \cos\theta c .}{1 + \frac{v \cdot \cos\theta \cdot c}{c^2}}$

$\cos\alpha = \frac{v_x}{c} = \frac{\sqrt{3}}{2}$

$\Rightarrow . \cos\theta = \frac{\cos\alpha - \beta .}{1 - \beta\cos\alpha} = 0.215$.

(2).

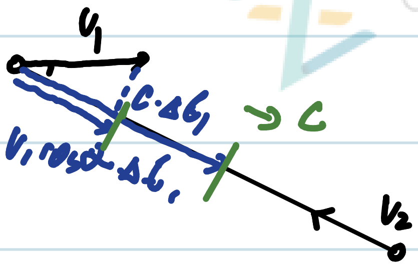


理解：① 光子.速度的方向.的反方向. ✓

② 光子.照亮区域方向 的反方向

地面系里. 直接带 非相对论多普勒.

发出间隔 $\Delta t_1 = \gamma_1 \Delta\tau_1$. 收到的间隔 $\Delta t_2 = \gamma_2 \cdot \Delta\tau_2$.



$\Delta t_2 = \frac{c\Delta t_1 - v_1\Delta t_1\cos\alpha}{c + v_2}$.

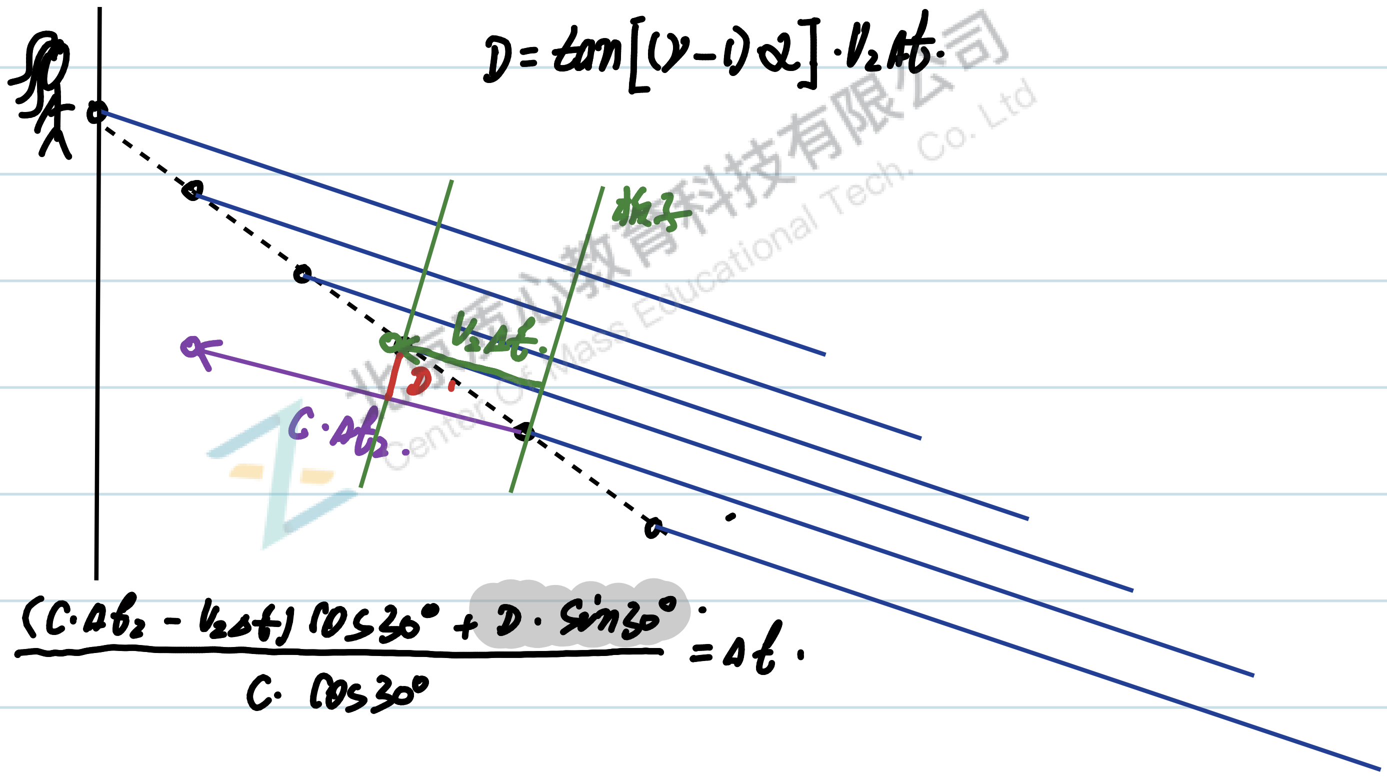
$\Rightarrow \Delta\tau_2 = \frac{1 - \beta_1\cos\alpha}{1 + \beta_2} \cdot \frac{\sqrt{1-\beta_1^2}}{\sqrt{1-\beta_2^2}} = \frac{5 - 2\sqrt{3}}{6}$.

(3).

反弹时间差. 在地面系. 仍为 Δt_2.

碰至在地面方向仍为 α 角

$D = \tan[(\gamma - 1)\alpha] \cdot v_2\Delta t$.



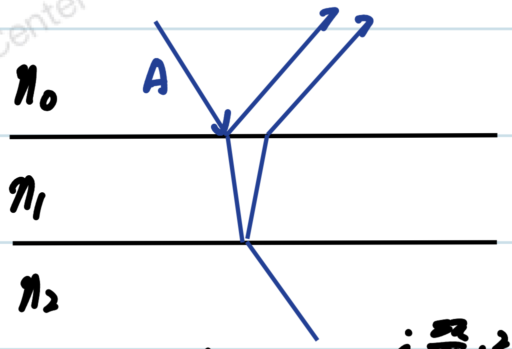
$\frac{(c \cdot \Delta t_2 - v_2\Delta t)\cos 30° + D \cdot \sin 30°}{c \cdot \cos 30°} = \Delta t$.

7. (1). 设. $r_1 = -\sqrt{R_1}$. $r_2 = -\sqrt{R_2}$.

光在. $n_0 \to n_1$. $n_1 \to n_0$. 透射振幅比. 为 $t_1 . t_1'$

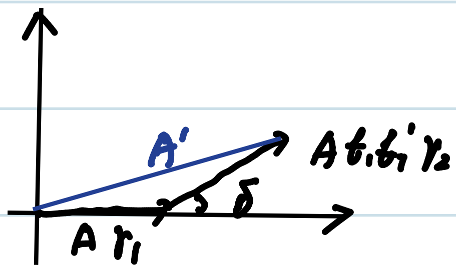
由. Stokes. 倒逆关系：$tt' - rr' = 1$.

入射光. A. $n_0 \to n_1$. $r_1 = -r_1' = -\sqrt{R_1}$, $n_1 \to n_2$. $r_2 = -r_2' = -\sqrt{R_2}$



$A' = A \cdot r_1 + A t_1 t_1' \cdot r_2 \cdot e^{i\frac{2\pi}{\lambda} \cdot 2n_1h}$

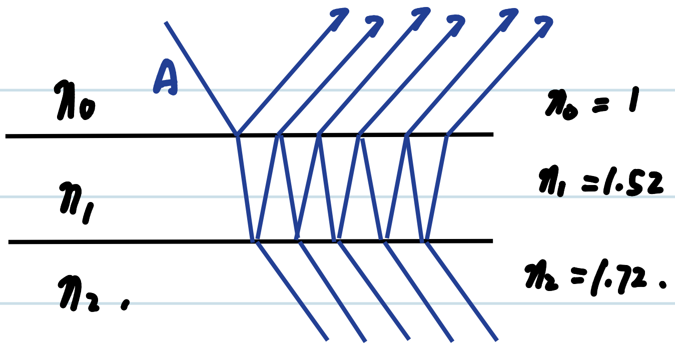
令 $\frac{2\pi}{\lambda} 2n_1h = \delta$.



$\Rightarrow I' = I_0 \cdot [r_1^2 + r_2^2(1 - r_1^2)^2 + 2 r_1 \cdot r_2 (1 - r_1^2)\cos\delta]$.

$R_1 . R_2 \ll 1$.

$\Rightarrow \frac{I}{I_0} = R_1 + R_2 + 2\sqrt{R_1R_2} \cdot \cos\delta$.



$$A'=Ar_1+At_1t_1'r_2e^{i\delta}\left(1+r_1'r_2e^{i\frac{2\pi}{\lambda}2n_1h}+(r_1'r_2e^{i\frac{2\pi}{\lambda}2n_1h})^2+\right.$$

$$\left.\cdots\cdots\right).$$

$$A'=Ar_1+A(1-r_1^2)r_2.e^{i\delta}\frac{1}{1-r_1'r_2e^{i\delta}}$$

$$=Ar_1+A(1-r_1^2)r_2.\frac{e^{i\delta}(1-r_1r_2\cos\theta+i\sin\theta\cdot r_1r_2)}{(1-r_1'r_2\cos\delta-ir_1'r_2\sin\theta)(1-r_1'r_2\cos\theta+ir_1'r_2\sin\theta)}.$$

$$=Ar_1+A(1-r_1^2)r_2\frac{(\cos\delta+i\sin\delta)(1-r_1'r_2\cos\theta+i\sin\theta\cdot r_1'r_2)}{(1-r_1'r_2\cos\delta)^2+r_1'r_2\sin^2\delta}$$

$$=A(a+bi).$$

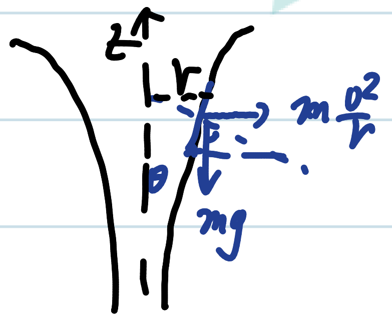
$$\frac{I'}{I_0}=a^2+b^2.$$ 若 r. r' 包含正负号. 则已包含半波损. 不用额外考虑.

8. $\vec{E}$. $\oint\vec{E}\cdot d\vec{s}=Q/\varepsilon_0$. $\oint\vec{E}\cdot d\vec{l}=\Phi_0$.

↓

$\vec{v}$. $\oint\vec{v}\cdot d\vec{s}=Q=\frac{\Delta V}{\Delta t}$. $\oint\vec{v}\cdot d\vec{l}=v\cdot2\pi r$.

水流速度 水管. 角动量



切向受力平衡.

$$mg\sin\theta=m\frac{v^2}{r}\cos\theta.$$

几何: $\tan\theta=\frac{dz}{dr}$

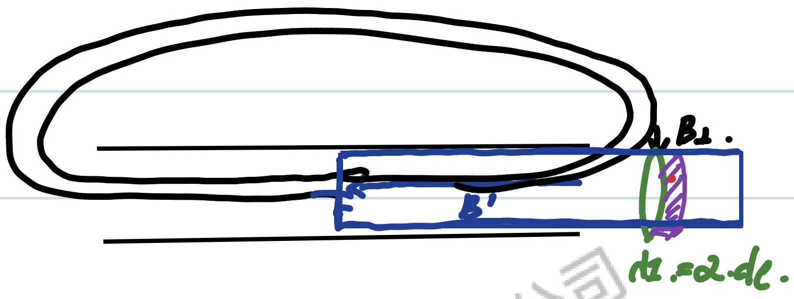
$$\Rightarrow\frac{dz}{dr}=\frac{v^2}{gr}.$$

$$\Rightarrow\frac{dz}{dr}=\frac{\Phi_0^2}{4\pi^2g}\cdot\frac{1}{r^3}$$

$$\Rightarrow z=z_0-\frac{\Phi_0^2}{8\pi^2g}\cdot\frac{1}{r^2}$$

取. $r\to\infty$. $z\to0$.

$$\Rightarrow z=-\frac{\Phi_0^2}{8\pi^2g}\cdot\frac{1}{r^2}$$



$F=\int B_1\cdot\alpha\,\underline{2\pi r\cdot dl}$. 线性磁化. 多个$\frac{1}{2}$.

$=\iint B_1\cdot ds\,\alpha$. α不是常数.

$=\frac{1}{2}\alpha\left(\oiint\vec{B}\cdot d\vec{s}-B\pi r^2\right)$ 是关于B. 关于l.

$=\frac{1}{2}B\cdot\pi r^2\alpha$. 的一次函数.

积分.有一个$\frac{1}{2}$.

$\alpha=M=\frac{J}{\mu_0}=(\mu_r-1)H.=(\mu_r-1)\frac{1}{\mu_0}B.$

$P=\chi\varepsilon_0E$. $J=\chi\mu_0H$.

$$\Rightarrow F=\frac{1}{2}(\mu_r-1)\mu_0n^2I^2\cdot\pi r^2.$$